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Project Title : Multi-Switch Flow Table Analyzer

```
sunny@LAPTOP-EB83KRGB:~/pox$ ./pox.py forwarding.l2_learning
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
```

POX controller is started and manages SDN switches.

```
sunny@LAPTOP-EB83KRGB:~$ sudo mn --topo linear,3 --controller=remote
[sudo] password for sunny:
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1, s1) (h2, s2) (h3, s3) (s2, s1) (s3, s2)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet>
```

Mininet topology with 3 switches (linear topology) is created.

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=16.5 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=7.89 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.150 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.185 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.219 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.092 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.157 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.160 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.141 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.167 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.215 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.159 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.158 ms
^C
--- 10.0.0.2 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12642ms
rtt min/avg/max/mdev = 0.092/2.011/16.453/4.646 ms
mininet>
```

Ping between h1 and h2 verifies connectivity and installs initial flow rules.

```
mininet> h1 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=24.3 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=1.44 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.171 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.099 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=1.30 ms
64 bytes from 10.0.0.3: icmp_seq=6 ttl=64 time=0.099 ms
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=0.182 ms
64 bytes from 10.0.0.3: icmp_seq=8 ttl=64 time=0.170 ms
64 bytes from 10.0.0.3: icmp_seq=9 ttl=64 time=0.170 ms
64 bytes from 10.0.0.3: icmp_seq=10 ttl=64 time=0.168 ms
64 bytes from 10.0.0.3: icmp_seq=11 ttl=64 time=0.153 ms
64 bytes from 10.0.0.3: icmp_seq=12 ttl=64 time=0.133 ms
64 bytes from 10.0.0.3: icmp_seq=13 ttl=64 time=0.177 ms
^C
--- 10.0.0.3 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12599ms
rtt min/avg/max/mdev = 0.099/2.198/24.325/6.402 ms
mininet>
```

Ping between h1 and h3 ensures traffic passes through multiple switches.

```
mininet> h3 ping h1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=12.0 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=2.30 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=2.01 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.170 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.177 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.122 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=0.162 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=0.252 ms
64 bytes from 10.0.0.1: icmp_seq=9 ttl=64 time=0.158 ms
64 bytes from 10.0.0.1: icmp_seq=10 ttl=64 time=0.202 ms
64 bytes from 10.0.0.1: icmp_seq=11 ttl=64 time=0.171 ms
^C
--- 10.0.0.1 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10505ms
rtt min/avg/max/mdev = 0.122/1.615/12.040/3.382 ms
mininet>
```

Reverse ping from h3 to h1 confirms bidirectional communication.

```
mininet> sh ovs-ofctl dump-flows s1
 cookie=0x0, duration=14.397s, table=0, n_packets=12, n_bytes=1176, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s1-eth1",vlan_tci=0x0000,
dl_src=b6:88:fe:c4:5e:fd,dl_dst=4a:db:9e:5a:d6:3a,nw_src=10.0.0.1,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth2"
 cookie=0x0, duration=14.391s, table=0, n_packets=12, n_bytes=1176, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s1-eth2",vlan_tci=0x0000,
dl_src=4a:db:9e:5a:d6:3a,dl_dst=b6:88:fe:c4:5e:fd,nw_src=10.0.0.2,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1"
 cookie=0x0, duration=9.239s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth1",vlan_tci=0x0000,dl_sr
c=b6:88:fe:c4:5e:fd,dl_dst=4a:db:9e:5a:d6:3a,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=1 actions=output:"s1-eth2"
 cookie=0x0, duration=9.237s, table=0, n_packets=2, n_bytes=84, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth2",vlan_tci=0x0000,dl_sr
c=4a:db:9e:5a:d6:3a,dl_dst=b6:88:fe:c4:5e:fd,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=1 actions=output:"s1-eth1"
 cookie=0x0, duration=9.235s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth1",vlan_tci=0x0000,dl_sr
c=b6:88:fe:c4:5e:fd,dl_dst=4a:db:9e:5a:d6:3a,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=2 actions=output:"s1-eth2"
 cookie=0x0, duration=9.230s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth2",vlan_tci=0x0000,dl_sr
c=4a:db:9e:5a:d6:3a,dl_dst=b6:88:fe:c4:5e:fd,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=2 actions=output:"s1-eth1"
mininet>
```

Flow table of switch s1 showing ICMP and ARP entries with packet counts.

```
mininet> h1 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=24.5 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=2.60 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.160 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.182 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.160 ms
64 bytes from 10.0.0.3: icmp_seq=6 ttl=64 time=0.129 ms
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=0.127 ms
64 bytes from 10.0.0.3: icmp_seq=8 ttl=64 time=0.128 ms
64 bytes from 10.0.0.3: icmp_seq=9 ttl=64 time=0.106 ms
64 bytes from 10.0.0.3: icmp_seq=10 ttl=64 time=0.156 ms
64 bytes from 10.0.0.3: icmp_seq=11 ttl=64 time=0.155 ms
^C
--- 10.0.0.3 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10504ms
rtt min/avg/max/mdev = 0.106/2.584/24.520/6.972 ms
mininet> sh ovs-ofctl dump-flows s2
 cookie=0x0, duration=18.261s, table=0, n_packets=11, n_bytes=1078, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s2-eth2",vlan_tci=0x0000,
dl_src=76:a3:33:4e:64:9d,dl_dst=c6:77:a1:f3:b5:9c,nw_src=10.0.0.1,nw_dst=10.0.0.3,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s2-eth3"
 cookie=0x0, duration=18.256s, table=0, n_packets=11, n_bytes=1078, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s2-eth3",vlan_tci=0x0000,
dl_src=c6:77:a1:f3:b5:9c,dl_dst=76:a3:33:4e:64:9d,nw_src=10.0.0.3,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s2-eth2"
mininet>
```

Flow table of switch s2 showing intermediate forwarding rules between switches.

```
mininet> h3 ping h1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=11.7 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=2.42 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.169 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.149 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.174 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.249 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=0.177 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=0.172 ms
64 bytes from 10.0.0.1: icmp_seq=9 ttl=64 time=0.193 ms
64 bytes from 10.0.0.1: icmp_seq=10 ttl=64 time=0.173 ms
^C
--- 10.0.0.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9483ms
rtt min/avg/max/mdev = 0.149/1.560/11.728/3.454 ms
mininet> sh ovs-ofctl dump-flows s3
 cookie=0x0, duration=14.029s, table=0, n_packets=10, n_bytes=980, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s3-eth1",vlan_tci=0x0000,d
l_src=06:5c:7f:49:5e:6c,dl_dst=b6:88:fe:c4:5e:fd,nw_src=10.0.0.3,nw_dst=10.0.0.1,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s3-eth2"
 cookie=0x0, duration=14.020s, table=0, n_packets=10, n_bytes=980, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s3-eth2",vlan_tci=0x0000,d
l_src=b6:88:fe:c4:5e:fd,dl_dst=06:5c:7f:49:5e:6c,nw_src=10.0.0.1,nw_dst=10.0.0.3,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s3-eth1"
 cookie=0x0, duration=8.987s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s3-eth1",vlan_tci=0x0000,dl_sr
c=06:5c:7f:49:5e:6c,dl_dst=b6:88:fe:c4:5e:fd,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=1 actions=output:"s3-eth2"
 cookie=0x0, duration=8.982s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s3-eth2",vlan_tci=0x0000,dl_sr
c=b6:88:fe:c4:5e:fd,dl_dst=06:5c:7f:49:5e:6c,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_op=1 actions=output:"s3-eth1"
 cookie=0x0, duration=8.981s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s3-eth1",vlan_tci=0x0000,dl_sr
c=06:5c:7f:49:5e:6c,dl_dst=b6:88:fe:c4:5e:fd,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,arp_op=2 actions=output:"s3-eth2"
 cookie=0x0, duration=8.971s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s3-eth2",vlan_tci=0x0000,dl_sr
c=b6:88:fe:c4:5e:fd,dl_dst=06:5c:7f:49:5e:6c,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,arp_op=2 actions=output:"s3-eth1"
mininet>
```

Flow table of switch s3 showing final forwarding behavior towards destination host.